Chapter 11

Radiation

Problem 11.1

Check that the retarded potentials of an oscillating dipole

$$V(r,\theta,t) = \frac{p_0\cos\theta}{4\pi\epsilon_0 r}\left\{-\frac{\omega}{c}\sin\left[\omega\left(t-\frac{r}{c}\right)\right] + \frac{1}{r}\cos\left[\omega\left(t-\frac{r}{c}\right)\right]\right\}$$
$$\boldsymbol{A}(r,\theta,t) = -\frac{\mu_0 p_0\omega}{4\pi r}\sin\left[\omega\left(t-\frac{r}{c}\right)\right]\hat{\boldsymbol{z}}$$

satisfy the Lorenz gauge condition.

Solution:

Problem 11.2

Problem 11.3

Problem 11.4

Problem 11.5

As the outlaws escape in their getaway car, which goes $\frac{3}{4}c$, the police officer fires a bullet from the pursuit car, which only goes $\frac{1}{2}c$. The muzzle velocity of the bullet (relative to the gun) is $\frac{1}{3}c$. Does the bullet reach its target (a) according to Galileo, (b) according to Einstein?

Solution: (a) Adding up the velocities with Galileo's velocity addition rule, the resulting velocity of the bullet is

$$\frac{1}{2}c + \frac{1}{3}c = \frac{5}{6}c > \frac{3}{4}c$$

so the bullet does reach its target.

Problem 11.6

Problem 11.7

Problem 11.8

Problem 11.9

Problem 11.10

Problem 11.11

Problem 11.12

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Problem 11.35

Problem 11.36